LupiScan

A project submitted
in partial fulfillment of the requirements for the degree of
Bachelor in Electrical Power Engineering

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Pledge

This is to declare that the project entitled “Project Title” is an original work done by the undersigned, in partial fulfillment of the requirements for the bachelor's degree at the ***Electric Power Engineering and Mechatronics department***, ***College of Engineering***, ***Tafila Technical University.***

All the analysis, design, and system development have been accomplished by the undersigned. Moreover, this project has **not** been submitted to any other college or university.

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CHAPTER 1: Introduction

This chapter introduces the development of a disease detection device aimed at identifying Lupus Fever using artificial intelligence and computer vision. The project is motivated by the need for a fast, cost-effective, and portable diagnostic solution. It addresses the challenges associated with traditional diagnostic methods by leveraging deep learning techniques implemented on a Raspberry Pi platform. This chapter also outlines the problem statement, objectives, scope, and structure of the report.

1.1 Overview

This project focuses on developing a disease detection device using artificial intelligence (AI) to identify Lupus Fever through facial image analysis. The device is designed using a Raspberry Pi 4, computer vision techniques, and deep learning (Convolutional Neural Networks) to accurately detect the disease. By leveraging AI and computer vision, the device aims to provide an efficient, low-cost, and portable solution for detecting Lupus Fever.

1.2 Project Motivation

The primary motivation for developing this project stems from the increasing need for rapid and accurate disease detection. Lupus Fever is often diagnosed late due to the absence of easily accessible diagnostic tools. This project aims to bridge this gap by utilizing AI to offer a quick, affordable, and portable diagnostic device. The uniqueness of this project lies in its integration of deep learning and edge computing on a compact platform, which provides real-time disease detection without needing an internet connection.

1.3 Problem Statement

The current diagnostic processes for Lupus Fever are often time-consuming, expensive, and require specialized laboratory equipment. There is a lack of portable diagnostic tools that can be used in remote or low-resource settings. This project addresses these issues by developing an AI-driven, cost-effective, and portable device capable of detecting Lupus Fever through facial image analysis.

1.4 Project Aim and Objectives

The primary goal of this project is to develop a portable AI-based device for the early detection of Lupus Fever. The objectives include:

1. Designing a device using Raspberry Pi 4 with integrated computer vision and deep learning capabilities.
2. Training a Convolutional Neural Network (CNN) model using Roboflow for accurate detection of Lupus Fever.
3. Conducting rigorous testing to ensure accuracy and efficiency.

1.5 Project Scope

The project scope includes designing and implementing a disease detection device that utilizes AI and computer vision to identify Lupus Fever. The device will be powered by a Raspberry Pi 4 and utilize YOLOv8 for image analysis. The project does not include the development of the mobile application itself, but the device will be compatible with mobile platforms for result transmission. The responsibility distribution involves the integration of hardware and software components, model training, and testing. Verification will involve accuracy assessment through a dataset obtained via RoboFlow and evaluated on a Raspberry Pi 4.

1.6 Project Software and Hardware Requirements

Table 1.1: The project requires software and hardware components to function effectively.

Hardware	Software
Raspberry Pi 4 (64-bit)	Raspbian OS (64-bit)
Camera module compatible with Raspberry Pi	Python 3.11.3
Power supply and necessary peripherals	Ultralytics , OpenCV
Wi-Fi and Bluetooth modules	YOLOv8 model trained on RoboFlow

1.7 Project Expected Output

The project aims to deliver a portable AI-powered device that can detect Lupus Fever through facial image analysis. The expected output is a clear indication of the presence or absence of the condition, displayed on a mobile device via Bluetooth or Wi-Fi communication.

1.8 Project Schedule

Table 1.2: Project Schedule.

Month	Project Schedule.
Month 1	Literature Review and Requirement Analysis.
Month 2	Hardware and Software Setup.
Month 3	Model Training, Testing, and Optimization.
Month 4	Device Integration, Final Testing, and Documentation.

1.9 Project, Product, and Schedule Risks

The project may encounter several risks that could potentially delay completion and affect the quality of the final product. These risks include:

- Hardware Failure: Raspberry Pi or camera module malfunction could delay integration and testing.
- Model Performance Issues: Low accuracy due to insufficient data quality may require retraining, leading to time loss.
- Compatibility Problems: Challenges integrating software libraries on the Raspberry Pi may delay development.
- Schedule Overruns: Unforeseen technical challenges could lead to extended testing or debugging phases.
- Resource Limitations: The Raspberry Pi's processing power may limit real-time performance, requiring model optimization.

Chapter 2: Literature Review

This chapter comprehensively reviews the research and technologies relevant to detecting Lupus Fever using machine learning and artificial intelligence. It encompasses the medical background of Systemic Lupus Erythematosus (SLE), the application of Convolutional Neural Networks (CNN) and transfer learning, and lupus diagnosis through machine learning.

2.1 Systemic Lupus Erythematosus

In the past decade, there has been significant research focused on Systemic Lupus Erythematosus (SLE), a chronic autoimmune disease that leads to severe health complications and is currently incurable. The Centers for Disease Control and Prevention (CDC) have reported that SLE is particularly prevalent among women aged 15-44, with approximately 200,000 adults diagnosed in the United States every year. The Lupus Foundation of America adds that about 90% of lupus patients are women, with SLE accounting for roughly 70% of all lupus cases [1].

Moreover, lupus is one of the top 20 leading causes of death. SLE triggers the immune system to attack healthy cells, causing organ damage, inflammation, and other serious complications [1],[2]. Despite the lack of a cure, early diagnosis can improve patient outcomes and extend their lifespan. According to the American College of Rheumatology, diagnosing SLE involves meeting at least 4 out of 11 criteria, including rash, photosensitivity, oral ulcers, non-erosive arthritis, pleuritis, renal disorders, neurologic disorders, hematologic disorders, immunologic disorders, and the presence of antinuclear antibodies [2]. However, the variability of these symptoms among patients makes diagnosis challenging. Traditional diagnostic methods, which include extensive blood tests including antinuclear antibody (ANA) test, urine tests, and biopsies, are time-consuming [2].

A significant indicator of SLE is the Butterfly Malar Rash (BMR), a distinctive butterfly-shaped rash across the cheeks and nose, observed in 46-75% of SLE patients. Diagnosing BMR requires skilled dermatologists due to its resemblance to other conditions like rosacea and fifth disease [3].

Skin diseases, particularly those linked to Systemic Lupus Erythematosus (SLE), frequently present with visible symptoms such as lesions, plaques, scaling, and pigmentation abnormalities. In SLE, the distinctive butterfly-shaped malar rash on the face is a common feature, often resulting in chronic pain, disfigurement, and severe emotional distress. The visibility of these conditions, especially on facial skin, which is constantly exposed and plays a critical role in personal identity and social perception, exacerbates their psychological impact [4], [5], [6]. Studies indicate that individuals with primary skin diseases, including SLE, psoriasis, and vitiligo, are significantly more likely to experience mental health issues such as anxiety and depression.

2.2 Convolutional Neural Networks and Transfer Learning

Convolutional Neural Networks (CNNs) are fundamental in medical image analysis due to their ability to detect spatial patterns effectively. For this project, we selected the YOLOv8 model, which is well-suited for object detection tasks, including identifying disease markers in facial images. YOLOv8's architecture allows for fast and accurate image analysis, making it ideal for deployment on edge devices like Raspberry Pi.

To optimize the model's performance, we utilized transfer learning by fine-tuning a pre-trained YOLOv8 model. This approach leverages the model's existing knowledge from large-scale datasets and adapts it to our specific use case. The model training and testing were conducted using the RoboFlow platform, which facilitated data annotation, augmentation, and model versioning.

2.3 Lupus Diagnosis Using Machine Learning

Machine learning techniques have been increasingly utilized to enhance lupus diagnosis accuracy. In this project, the primary focus was on detecting the butterfly-shaped rash indicative of Lupus Fever. Given the limited availability of high-quality images depicting this rash, data augmentation played a vital role in training the model effectively.

Data Augmentation Techniques

Due to the scarcity of images depicting the butterfly rash, we employed several augmentation techniques to artificially increase the size and diversity of the training dataset. These methods included:

1. **Flipping:** Horizontally flipping images to introduce variations in facial symmetry.
2. **Rotation:** Adjusting the angle of the image to account for head tilt and facial orientation differences.
3. **90-Degree Rotation:** Rotating images by 90 degrees to simulate different facial orientations.
4. **Brightness Adjustment:** Modifying the brightness to account for variations in lighting conditions.
5. **Blur:** Adding slight blurring to enhance robustness against low-quality images.
6. **Saturation and Hue Adjustment:** Modifying color intensity and hue to represent different skin tones and lighting.

These augmentation techniques were applied using the RoboFlow platform, which streamlined the process and allowed for consistent and controlled variation. By augmenting the dataset, we significantly enhanced the model’s ability to generalize and detect the rash under diverse real-world conditions.

Chapter 3: Requirement Engineering and Analysis

3.1 System Description

The primary goal of this project is to develop a system that utilizes computer vision and machine learning to detect Lupus Fever by identifying the characteristic butterfly-shaped rash on the face. The system leverages a Raspberry Pi equipped with a camera to capture facial images, which are then processed using a trained deep learning model to identify the presence of the rash. The output is sent to a mobile application via Wi-Fi or Bluetooth for real-time monitoring.

3.2 Software and Hardware Requirements

This section outlines the required software and hardware components necessary for the implementation of the system.

3.2.1 Software Requirements

- **Operating System:** Raspberry Pi OS (32-bit)
- **Programming Languages:** Python 3
- **Libraries:** Ultralytics, OpenCV, NumPy
- **Platform:** RoboFlow for dataset preparation and augmentation
- **Development Environment:** Visual Studio Code, Jupyter Notebook
- **Communication Protocols:** Wi-Fi, Bluetooth

3.2.2 Hardware Requirements

- **Processing Unit:** Raspberry Pi 4 Model (64-bit)
- **Camera Module:** Raspberry Pi Camera v2.1
- **Power Supply:** 5V, 3A USB-C adapter
- **Connectivity:** Wi-Fi module, Bluetooth adapter
- **Display:** HDMI or LCD screen for local monitoring

3.3 Project Limitations

The primary limitations of the project are:

- **Data Scarcity:** A Limited number of labeled images of the butterfly rash, requiring augmentation.
- **Hardware Constraints:** Processing limitations due to the Raspberry Pi's capabilities.
- **Environmental Variability:** Accuracy may decrease under poor lighting or unconventional camera angles.
- **Model Generalization:** The model is specifically trained for butterfly rash detection and may not detect other lupus manifestations.

3.4 Project Expected Output

The expected output of the project is a system capable of detecting the characteristic butterfly rash associated with Lupus Fever from facial images. The system should achieve a high accuracy rate (targeting above 80%) and provide real-time feedback through a mobile application. The system should also be lightweight and efficient enough for deployment on a Raspberry Pi.

Chapter 4: Architecture and Design

4.1 Overview

This chapter details the system's architecture, including the software and hardware components, to ensure efficient and accurate detection of lupus rash.

4.2 Software Architecture

The software architecture consists of multiple layers, each responsible for a specific function, from data acquisition to prediction and result transmission.

4.2.1 Logical View

The system is logically divided into the following components:

- **Data Acquisition:** Capturing facial images using the camera.
- **Preprocessing:** Applying augmentation and normalization.
- **Inference:** Utilizing the YOLOv8 model for rash detection.

4.2.2 Process View

The data flow in the system follows these steps:

1. Capture an image from the camera.
2. Preprocess the image (resizing, normalization).
3. Run the YOLOv8 model for rash detection.
4. Generate a probability score and bounding boxes for lupus rash.

4.2.3 Physical View

The physical setup includes the Raspberry Pi 4 connected to a camera module, with a power supply and wireless communication interfaces for data transmission.

4.2.4 Component Details

- **Camera Module:** Captures facial images.
- **Processor (Raspberry Pi):** Runs the trained YOLOv8 model.

4.3 Software Design

The design follows a modular approach, allowing each component to function independently. The image preprocessing and model inference are separate from the data transmission module, enabling flexibility and scalability.

Chapter 5: Implementation and Testing

5.1 Description of Implementation

The implementation involved setting up the Raspberry Pi 4, integrating the camera, and loading the trained YOLOv8 model. The system was programmed to capture images, preprocess them, and perform inference using the trained model. The primary focus was to ensure that the system could accurately detect the butterfly-shaped rash indicative of Lupus Fever in real-time.

Training was conducted on Roboflow's model training environment, utilizing augmentation techniques such as flipping, rotation, brightness adjustment, blur, saturation, hue, and 90-degree rotation. The augmented dataset increased from 106 original images to a total of 610 images, significantly improving the model's robustness.

5.2 Programming Language and Technology

- **Python 3:** Primary programming language
- **Ultralytics:** Running the deep learning model on the Raspberry Pi
- **OpenCV:** Image capture and preprocessing
- **Roboflow:** Dataset management and augmentation
- **YOLOv8:** The primary model used for detecting lupus manifestations

5.3 Implementation Snippets

```python
import cv2
import numpy as np
import subprocess
from ultralytics import YOLO
# Load YOLOv8 model
model = YOLO("/home/rama/yolovenv/yolo/best.pt")
# Frame size
WIDTH = 640
HEIGHT = 480
# Start libcamera-vid stream (raw YUV420 output to stdout)
cmd = [
    "libcamera-vid",
    "-t", "0",  # unlimited time
    "--width", str(WIDTH),
    "--height", str(HEIGHT),
    "-o", "-",  # output to stdout
    "--codec", "yuv420"]

# Start subprocess
```

```
p = subprocess.Popen(cmd, stdout=subprocess.PIPE,
bufsize=WIDTH * HEIGHT * 3)
while True:
    # Read one YUV420 frame (1.5 bytes per pixel)
    raw_frame = p.stdout.read(int(WIDTH * HEIGHT * 1.5))
    if not raw_frame:
        print("Failed to grab frame")
        break
    # Decode YUV420 to BGR (for OpenCV)
    yuv = np.frombuffer(raw_frame,
dtype=np.uint8).reshape((int(HEIGHT * 1.5), WIDTH))
    frame = cv2.cvtColor(yuv, cv2.COLOR_YUV2BGR_I420)
    # Convert to RGB (YOLO expects RGB input)
    rgb_frame = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
    # Run YOLO detection
    results = model(rgb_frame)
    # Draw results on the original frame
    for r in results:
        for box in r.boxes:
            x1, y1, x2, y2 = map(int, box.xyxy[0])
            cls = int(box.cls[0])
            conf = float(box.conf[0])
            label = model.names[cls]
            cv2.rectangle(frame, (x1, y1), (x2, y2), (0, 255,
0), 2)
            cv2.putText(frame, f"{label} {conf:.2f}", (x1, y1
- 10),
                        cv2.FONT_HERSHEY_SIMPLEX, 0.5, (0,
255, 0), 2)
    # Display the frame
    cv2.imshow("YOLOv8 Detection", frame)
    # Exit when pressing 'q'
    if cv2.waitKey(1) & 0xFF == ord('q'):
        break
# Terminate subprocess and close window
p.terminate()
cv2.destroyAllWindows()
```

Figure 5.1: Code Script.

5.4 Testing and Results

Testing involved evaluating the model on the augmented dataset as well as real-time facial images captured via the Raspberry Pi camera. The following metrics were used:

Table 5.1: The Metrics of Model Yolov8 that were used in the Project

Model	mAP	Precision	Recall
Yolov8	84.2%	91.4%	73.2%

Figure 5.2: Accurately Detect the Butterfly-Shaped Rash for The Face.

Figure 5.3: Accurately Detect the Butterfly-Shaped Rash for Side Face.

Figure 5.4: Accurately Detect the Butterfly-Shaped Rash for the Ear.

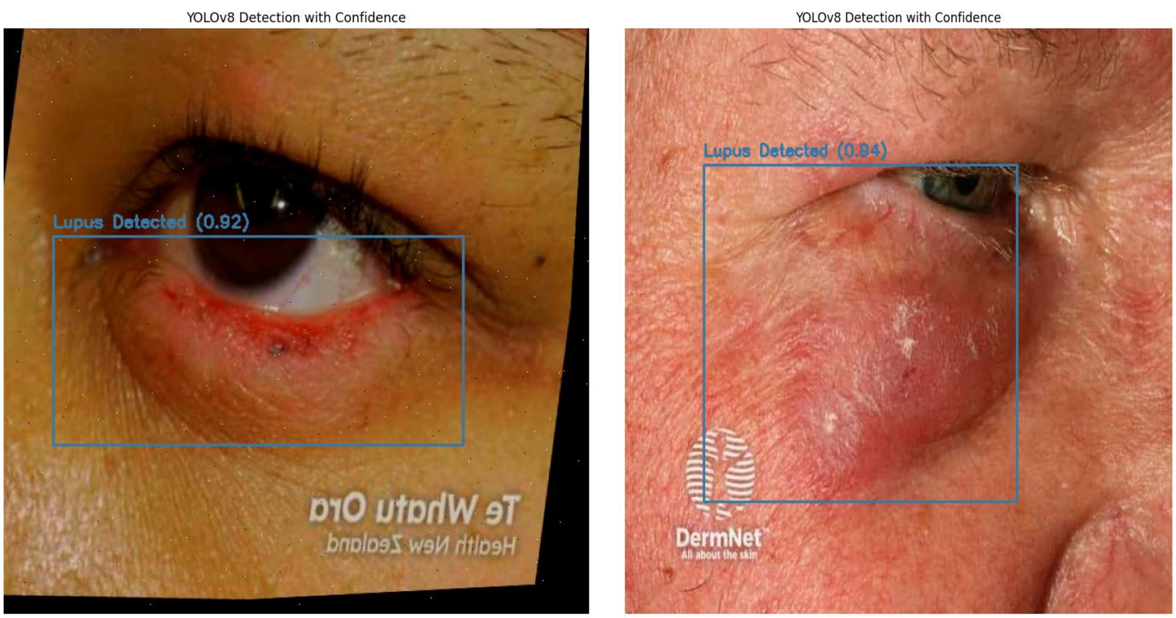

Figure 5.5: Accurately Detect the Butterfly-Shaped Rash for the Eye.

Figure 5.6: Accurately Detect the Butterfly-Shaped Rash for the Lips.

Chapter 6: Conclusion and Results

The project successfully developed a lightweight system for detecting the butterfly rash associated with Lupus Fever using the YOLOv8 model. The model's high accuracy and precision make it a reliable tool for detecting lupus manifestations in real-time on a Raspberry Pi.

Key Findings

- The model was trained on three different object detection models: RF-DETR, YOLOv8, and YOLOv11. Among them, YOLOv8 demonstrated the highest accuracy and precision, making it the most suitable for this application.
- YOLOv8 outperformed the other models in both precision and mAP, proving to be the most effective choice for medical imaging tasks related to lupus detection.
- Data augmentation played a crucial role in improving model accuracy, addressing the challenge of limited data availability.

Table 6.1: The Metrics of Three Different Object Detection Models.

Model	mAP	Precision	Recall
RF-DETR	82.9%	100%	100%
Yolov11	82.9%	89.8%	75.3%
Yolov8	**84.2%**	**91.4%**	**73.2%**

Recommendations

- Increasing the dataset size by incorporating more diverse images could further enhance accuracy.
- Implementing more advanced augmentation techniques may improve generalization in varied clinical settings.
- Exploring interpretability methods like Grad-CAM could provide insights into the model's decision-making process.

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